

Midterm 总结

一、8 Golden Rule

1. Describe Three design flaws with the interface.
2. Name two of the Eight Golden Rules that have been violated by this interface. Give a brief explanation of how they were violated.

一、8 Golden Rule Q1: 8 Golden rule - 3 different parts

Examine the following interface.

(2) Label is not at right

(3) 用checkbox,而不是radio button

1. Describe FOUR design flaws with the interface. (8 marks) 1. Three design flaws - 同样问法

出现文本框不对齐的现象 (a->b):

- (1).
 - a. Text boxes are not aligned (垂直/水平方向 -> 开头对齐)
 - b. Inconsistent label placements 标签位置不一致

出现日期, 大概率:

- c. DOB text box does not prevent errors -> 用户可以输入无效日期
- d. Date format is unknown -> 用户不知道应该输入哪种格式 (如 2025/11/11、11-11-2025 或 11月11日)

是否有cancel? A Cancel/Clear button is not available

2. Which golden rule this interface violate

2. Name two of the Eight Golden Rules that have been violated by this interface. Give a brief explanation of how they were violated. (12 marks, 6 marks each) 看到Date-> prevent error -> 用date picker

文本框不对齐:

Strive for consistency --> The text boxes are not aligned properly.

出现日期: 100%会出现

Prevent errors -->The DOB box does not prevent errors; incorrect date could be entered and accepted. 2-D menus(Date picker) should be used to replace textboxes.

RADIO BUTTONS	CHECKBOXES
☐ _____	☐ _____
☒ _____	☒ _____
☐ _____	☐ _____
☐ _____	☒ _____

二、Direct Manipulation (25)

1. **Explain the principles of direct manipulation.**(15)
2. **Name** two ways you could update the interface in Question 1.1 to support these principles. Draw a sketch of your redesign below **【记得带铅笔和尺子】**

[2023]

15分

1. **Explain the principles of direct manipulation.**(10 marks) **Q2**
 - Continuous representations of the objects and actions of interest with meaningful **visual metaphors**.
 - Physical actions or presses of labeled buttons, instead of complex syntax.
 - Rapid, incremental, **reversible actions** whose effects on the objects of interest are visible immediately.

10分

2. **Name two ways you could update the interface in Question 11 to support these principles. Draw a sketch of your redesign below.**(10 marks)
 - 1 -> A 2D menu should be used to let the user select a date **Date /Cancel button的问题会出现在考卷**
 - 3 -> A CANCEL/CLEAR button should be available

- **Continuous** representations of the objects and actions of interest with meaningful visual metaphors 隐喻.

用连续可视化的方式呈现用户关注的对象和操作，并通过有意义的视觉隐喻来增强理解

- **Physical actions** or **presses of labeled buttons**, instead of complex syntax.

强调“可操作性”和“可识别性”，用户不需要记住命令，只需点击明确标识的按钮。

- **Rapid, incremental, reversible** actions whose effects on the objects of interest are **visible immediately**.

三、Expert review v.s. usability testing (20)

1. Meaning

An expert review is a method where **3~5 experts** evaluate a product, service, or system to identify **potential problems and suggest improvements**.

Usability testing is a method of evaluating a product like a website or app by **observing real users as they attempt to complete tasks with it**

2. When?
3. Weakness(Limitation)?

	Usability Testing	Expert Reviews
Perspective	Real users performing real tasks User-centered 真实验证用户行为，发现实际使用中的障碍	Experts applying heuristics and experience flexible and iterative 快速识别界面问题，基于专家经验与设计原则
Strengths	Reveals actual user behavior and pain points	Identifies design rule violations and consistency issues
<u>Weakness</u>	- Time-consuming (recruitment, testing, analysis) - Higher cost - Requires skilled facilitation	- May miss context-specific behaviors 无法发现那些只有在真实操作中才会暴露的问题 - Subjective: depends on expert quality - Risk of superficial analysis Limited availability of experts with both domain and usability expertise (hard to recruit)
Cost	Slower	Faster
Speed	more expensive	cheaper
Depth of Insight	Rich behavioral data , emotional reactions	Systematic, rule-based diagnosis
Best For	Validating usability with target audience	Early-stage design critique or budget-limited projects
<u>When?</u> 	Only Testing	Every phase

When Usability Testing Is Better?

- **Need real user feedback** on task completion, confusion points, or emotional reactions.
- **Testing a live or high-fidelity prototype** and need to validate assumptions.
- **Targeting specific user groups** (e.g., elderly, children, professionals) whose behavior may differ from expert expectations.

When Expert Reviews Are Better?

- In **early design stages** and want fast, low-cost feedback.
- **Need to catch heuristic violations** like inconsistent layouts
- **Under time or budget constraints** and can't recruit users.

4. 顺序的重要性 (ER->UT)

Efficiency	Expert Review helps eliminate obvious design flaws before testing, saving time and cost .
Focus	It allows usability testing to concentrate on deeper behavioral issues , not surface-level mistakes.
User Experience	Prevents users from getting stuck on basic problems, improving the quality of feedback.
Validation Loop	Expert assumptions can be validated through user behavior , creating a structured evaluation cycle.

Expert Review first ensures the interface is clean enough for meaningful testing.

Usability Testing second provides real-world validation and behavioral insights.

Reversing the order risks wasting resources and collecting low-quality data.

Risks of Reversing the Order

- Testing sessions may focus on trivial problems. 测试过程可能集中在琐碎问题上，无法深入挖掘真正的可用性障碍
- You may **spend more time** fixing things that experts could've flagged earlier.

5. 3~5 EXPERT 原因

Using 3–5 experts ensures a rich, reliable, and cost-effective evaluation. It balances depth of insight with efficiency, making it the optimal range for most UX review scenarios.

【拓展：from Hint】

Give two **reasons** why **expert reviews are useful**. Also give two **limitations** of expert reviews.

Reasons :

a. Identify Design Flaws Early

Experts can spot usability issues, inconsistencies, and violations of design principles **_before the product reaches users.**

This helps teams **save time** and **cost** by **avoiding rework** after deployment.

b. Leverage Domain Knowledge and Standards

Experts apply **established heuristics, guidelines, and best practices** to evaluate interfaces systematically.

Their feedback is grounded in **prior experience and research**, making it more reliable than casual user impressions.

Limitations:

The dangers with expert reviews are that the experts may **not** have an adequate understanding of the task domain or user communities.

Even experienced expert reviewers have great **difficulty** knowing how typical users, especially first-time users, will really behave

An **expert review** is a method of evaluating a product, system, or application by having one or more subject matter **experts use their knowledge** to **identify** potential problems from a user's perspective

专家在进行界面评估或可用性分析时，**核心职责是从用户角度出发**，但他们也必须**理解并权衡投资人和利益相关者的目标**

“**Heuristic**” 启发式原则：一种基于经验的评估标准，用于判断界面设计是否合理、易用、符合用户心理模型。

“**First-time user**” 是指第一次使用某个系统、软件或界面的人。

四、Expert review (Think aloud protocol)

1. Difference
2. Meaning

Think-aloud protocol is a technique used in usability testing and cognitive research where users are asked to ***say out loud what they are thinking*** as they perform tasks.

The **concurrent think-aloud protocol** is a usability testing method where participants **verbally narrate** their thoughts, feelings, and actions **while performing a task**

Retrospective Think-Aloud (RTA) is a usability testing method where participants **complete** a task and then, after the session, verbally **explain** their thoughts, feelings, and reasoning while reviewing a recording of their own performance

	Concurrent Think–Aloud	Retrospective Think–Aloud
Timing	During task	After task
Data freshness	Real–time	Memory–based
Task interference	Possible	None
Common use	Early–stage usability testing	Post–task analysis or interviews

3. Limitation

Concurrent Think–Aloud (CTA)

用户在执行任务的同时 verbalize（说出）自己的想法

Cognitive overload	Speaking while performing tasks increases mental effort , which may affect task performance.	同时操作和表达思维会增加认知负荷，可能影响任务表现。
Behavioral distortion	Verbalizing thoughts may alter natural decision–making , leading to less authentic behavior.	说话可能改变用户的自然决策路径，导致行为不真实。
Surface–level insights	Users may only express immediate thoughts, lacking deeper reflection or emotional context.	用户可能只表达表层想法，缺乏深度反思或情绪背景。

Practical Limitations of CTA – 对环境资源的要求

Usability laboratory with one–way mirrors	带有单向镜的可用性实验室	允许观察者在不打扰用户的情况下观察其行为
Multiple cameras / Digital video cameras	多个摄像头 / 数码摄像机	从不同角度记录用户操作过程和面部表情
Intercom	对讲系统	测试人员与观察者之间沟通，或为用户互动

Requires controlled environment	CTA often needs a quiet, distraction–free space to ensure users can speak and perform tasks naturally.	通常需要安静、无干扰的环境，确保用户能自然地操作并 verbalize。
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Needs one-way mirror or observation setup	For professional-grade CTA, usability labs may use one-way mirrors or live observation rooms , which are costly .	专业测试常用单面镜或观察室，成本高昂，小公司难以负担。
Demands skilled facilitation	Facilitators must guide without interfering, which requires training and experience.	引导者需具备专业技巧，避免干扰用户行为，培训成本高。
Not scalable for remote or agile teams	In fast-paced or remote settings, setting up CTA sessions may be impractical.	对于远程或敏捷团队，组织 CTA 测试难度大，不易扩展。

Retrospective Think-Aloud (RTA)

用户在完成任务后 verbalize（回顾）自己的思维过程，通常配合视频回放

Memory bias	Users may forget or misrepresent their actual thoughts and motivations.	用户可能忘记或误解当时的真实想法与动机。
Rationalization	Users tend to reconstruct their behavior logically , which may not reflect real-time decisions.	用户倾向于“合理化”行为，而非真实还原当时的决策。

Why **Concurrent Think-Aloud** Is Often **Better**?

Captures real-time cognition	CTA records users' thoughts as they happen, providing direct access to their decision-making process.
Reveals immediate confusion or hesitation	You can observe where users struggle, hesitate, or express uncertainty in the moment, which is often lost in retrospective recall.
Minimizes memory distortion	Since users verbalize during the task, there's no reliance on memory, reducing the risk of rationalization or forgetting.
More authentic reactions	Emotional responses (e.g., frustration, surprise) are captured live, offering richer usability insights.
Supports task-context linkage	Observers can directly correlate user comments with specific interface elements or actions, improving diagnostic accuracy.

Step for UT

1. **Plan and prepare:** Define goals, Create the test plan, Recruit participants, Prepare the environment, Run a pilot test
 2. **Conduct the test:** Welcome and brief the participant, Facilitate the session, Collect data
 3. **Analyze and report:** Analyze findings, report results, follow up
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五、Gulf (10)

What is gulf of execution? What is gulf of evaluation? Give an example of each.(Ch3)

- **Gulf of execution** 执行鸿沟: Mismatch between the **user's** intentions and the allowable actions 用户想做某事，但界面不支持或不清楚怎么做。
 - 例如：用户想保存文档，但找不到“保存”按钮。
- **Gulf of evaluation** 评估鸿沟: Mismatch between the **system's** representation and the user's expectations 系统的反馈方式与用户的预期不一致。
- 给 Case 判断，并解释原因。

用户预期	系统反馈	结果
红色表示错误	系统用绿色显示“错误已处理”	用户以为操作成功
点击“保存”后无提示	页面静默刷新	用户不确定是否保存成功